

MATH8009: Linear Algebra - Problem Sheet 9 Answers

$$\text{Q1 a) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \quad \text{b) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{46}{27} \\ -\frac{13}{27} \end{pmatrix} \quad \text{c) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 41 \\ -17 \end{pmatrix} \quad \text{d) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Q2 See Q1

$$\text{Q3 a) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{11}{6} \end{pmatrix} \quad \text{b) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ -\frac{9}{2} \end{pmatrix} \quad \text{c) } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{2} \end{pmatrix}$$

$$\text{Q4 a) } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \text{b) } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix} \quad \text{c) } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ -2 \end{pmatrix} \quad \text{d) } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$$

Q5

$$\text{a) } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix}$$

$$\text{b) } x_3 \text{ is free} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 - 3t \\ 1 - 4t \\ t \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$\text{c) } x_2 \text{ is free} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 - 6t \\ t \\ 4 \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$\text{d) } x_3 \text{ is free} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 6 - 3t \\ 3 - 4t \\ t \\ -5 \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$\text{e) } x_2 \text{ is free} \rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 8 - 2t \\ t \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$\text{f) } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

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g) x_4 is free $\rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -1 - 4t \\ 6 - 2t \\ 8 - 3t \\ t \end{pmatrix}$ for some $t \in \mathbb{R}$

h) x_2 and x_5 are free $\rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -2 - 3s - 4t \\ s \\ 1 + 2t \\ 3 - 5t \\ t \end{pmatrix}$ for some $s, t \in \mathbb{R}$